

- Place dots on the coil terminals in figure 1 to obtain correct dot conventions for the coupled coils.

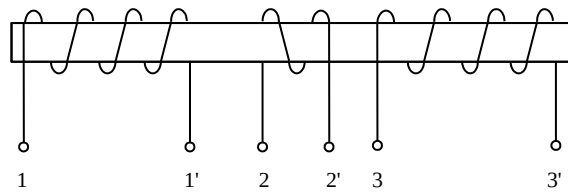


Figure 1

- Write loop equations in both time and phasor domain for the circuits in figure 2.

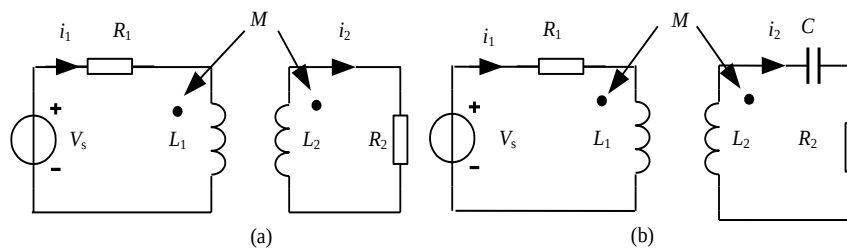


Figure 2

- For the circuit of figure 3, determine the value of angular frequency ω such that the impedance $Z = r$. Take $L_1=0.01\text{H}$, $L_2=0.02\text{H}$, $C=2\mu\text{F}$ and $M=0.01\text{H}$.

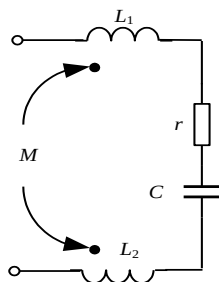


Figure 3

- Find input impedance Z_{ab} at angular frequency ω for each circuit in figure 4.

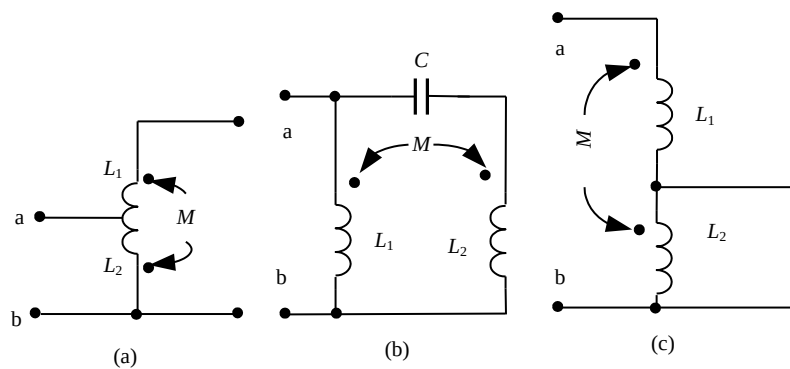


Figure 4

- Find $\dot{V}_2$ for the circuit in figure 5. Take $k=1/2$.

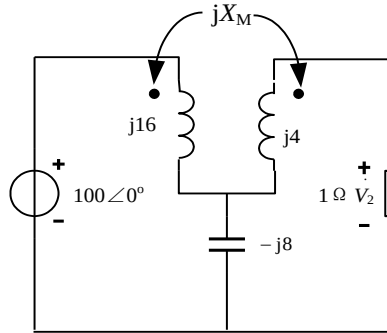


Figure 5

6. Find $\dot{I}_1$ in figure 6. Take $r = 10\Omega$, $L_1 = 4 \text{ mH}$, $L_2 = 6 \text{ mH}$, $M = 3 \text{ mH}$, $L_3 = 10 \text{ mH}$, $C = 100 \mu\text{F}$ and $v_s = 100\sqrt{2} \sin(1000t + 60^\circ)(\text{V})$.

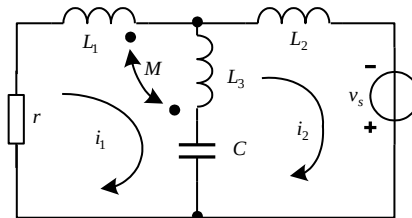


Figure 6

7. Determine the equivalent inductance for the input impedance for the circuit in figure 7.

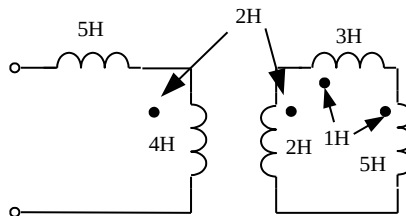


Figure 7

8. Find the input impedance Z_i and the current phasor $\dot{I}_2$ in figure 8.

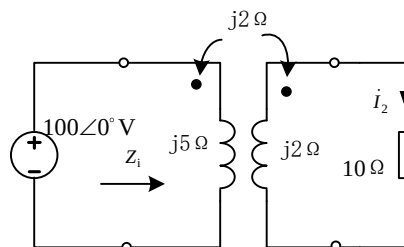


Figure 8

9. Find the power received by the resistor in secondary loop of the circuit in figure 9.

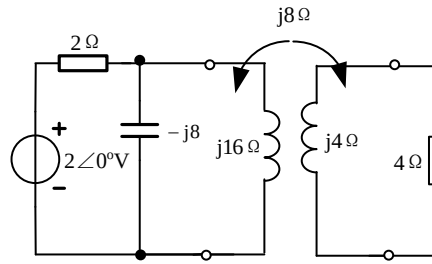


Figure 9

10. Determine the Thevenin equivalent circuit of the circuit in Fig. 10 to the left of the terminals.

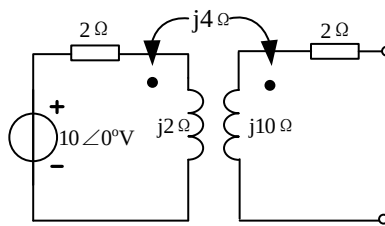


Figure 10

11. For the circuit in figure 11, determine the load impedance Z_L that draws maximum average power and find the power under the following restrictions to the load:

- (1) Z_L is a general impedance with resistive and reactive part.
- (2) Z_L is a resistor.

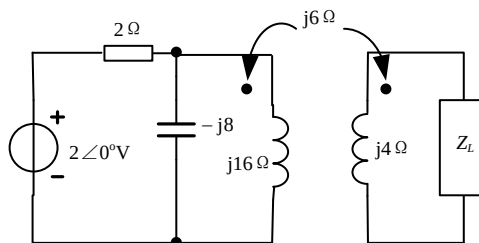


Figure 11

12. For the circuit in figure 12, determine the values of X_c and n such that the load Z receive maximum average power, and find this maximum power.

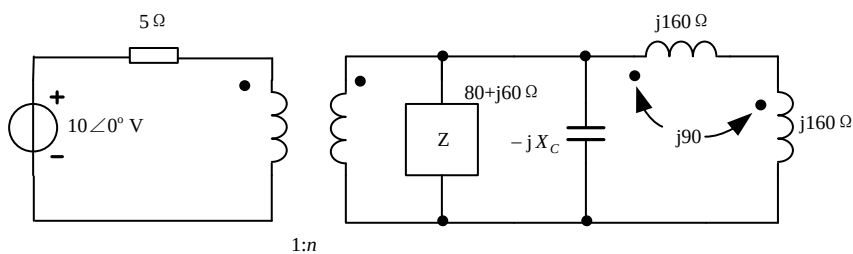


Figure 12

13. Calculate the average powers supplied by the source and absorbed by secondary load of the circuit in figure 13.

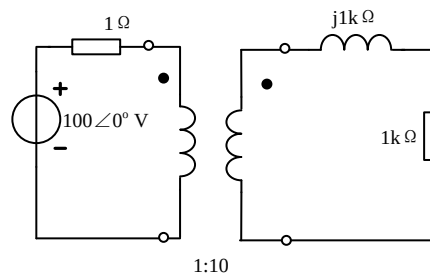


Figure 13

14. For the circuit in figure 14, find the turn ratio n of the matching transformer and the maximum average power drawn by the load R_L . Take $R_s=10k\ \Omega$, $R_L=3\ \Omega$ and $\dot{V}_s = 100\angle 0^\circ\text{ V}$.

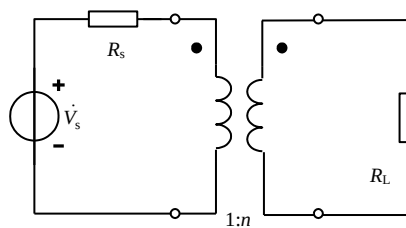


Figure 14

15. Find the currents through all elements in the circuit of figure 15.

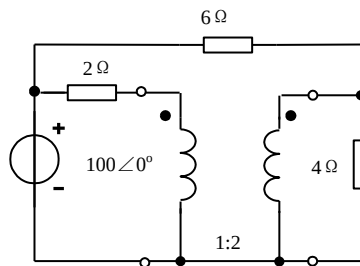


Figure 15

16. Determine the input impedance to the right of the terminals of the circuit in figure 16.

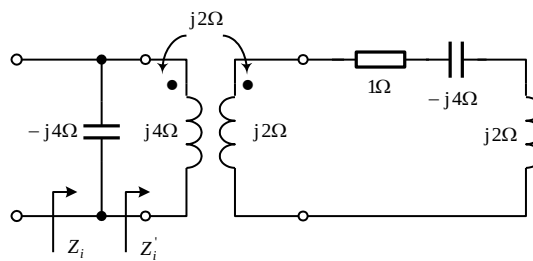


Figure 16

17. Determine the turn ration n , the maximum average power drawn by R_L and the current of the secondary loop of the circuit in figure 17. Take $\dot{V}_s = 100\angle 0^\circ\text{ V}$, $\dot{I}_s = 100\angle 0^\circ\text{ A}$, $R_1=R_2=1\ \Omega$ and $R_L=10\ \Omega$.

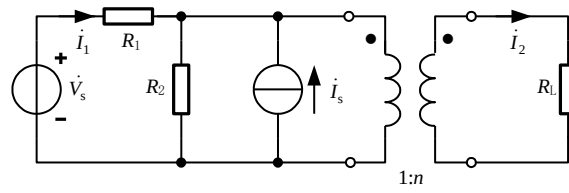


Figure 17

18. Find the input impedance for the circuit in figure 18. Take $n=0.5$.

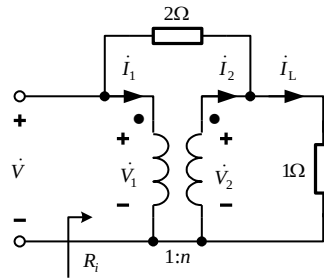


Figure 18

19. As shown in figure 19(a), the Thevenin equivalent of an audio signal source is connected to a load speaker of 8Ω .

- (1) Find the power drawn by the speaker, and show that this power is lower than the maximum power that a speaker could absorb from the source.
- (2) Insert a matching transformer between the source and the speaker to achieve maximum average power transfer to the speaker as shown in figure 19(b). Determine the turn ratio n , the input impedance seen by the source to the right and the maximum power drawn by the speaker.

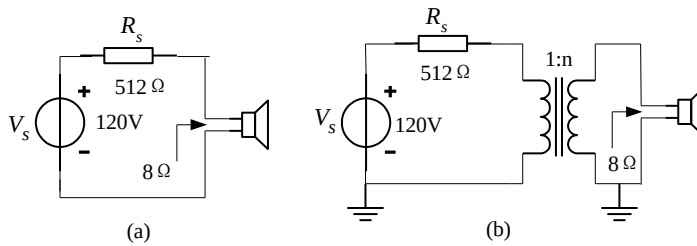


Figure 19